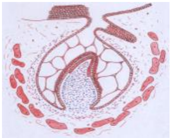

EARLY TOOTH DEVELOPMENT

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BDS,MDS

Assistant Professor

Oral Biology

- Where do teeth come from?
- How does a soft epithelial thickening become a hard tooth?
- What happens if something goes wrong during development?

Oral ectoderm → Dental lamina → Bud → Cap

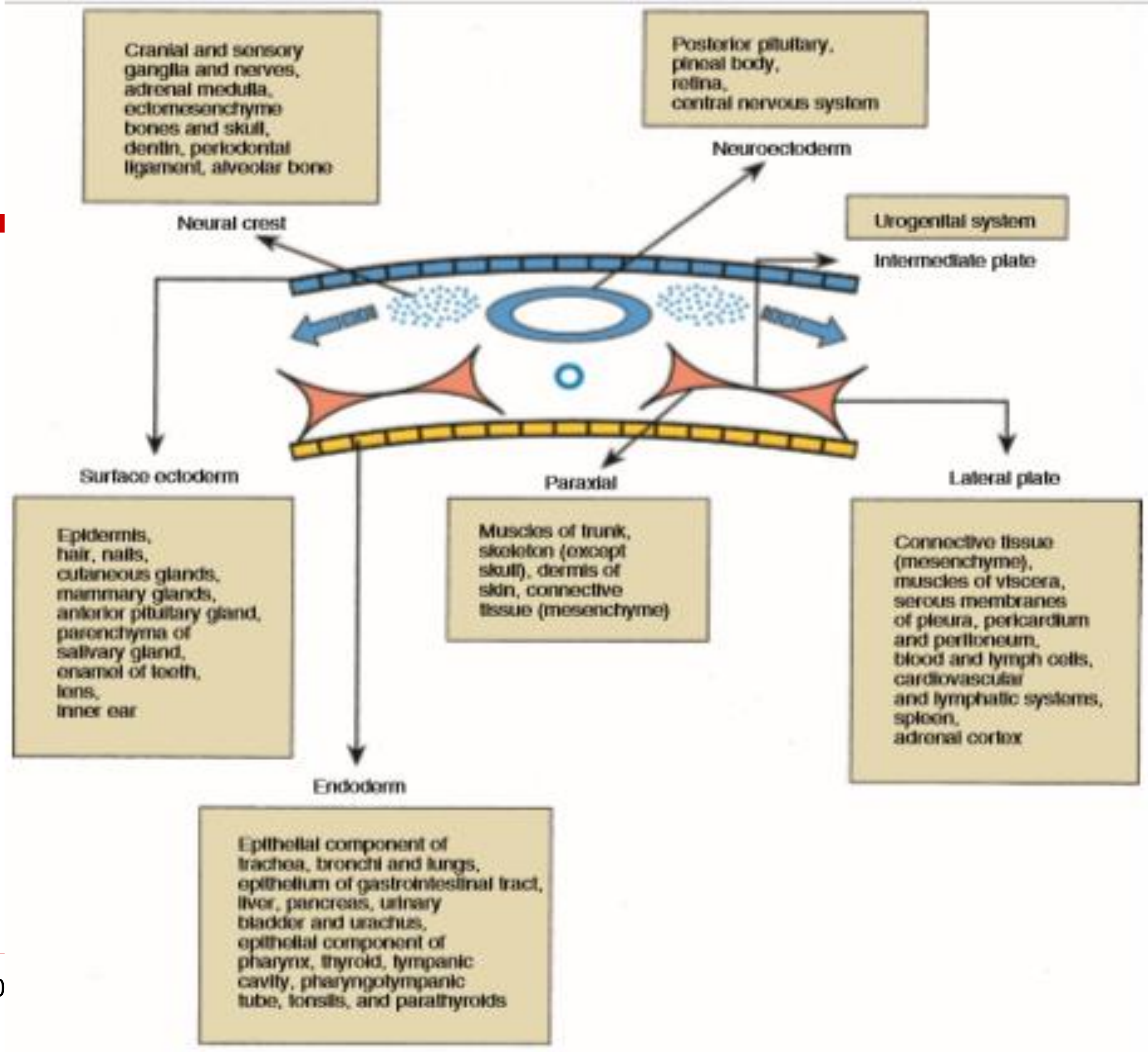
→ Bell → Hard tissue formation → Root formation →

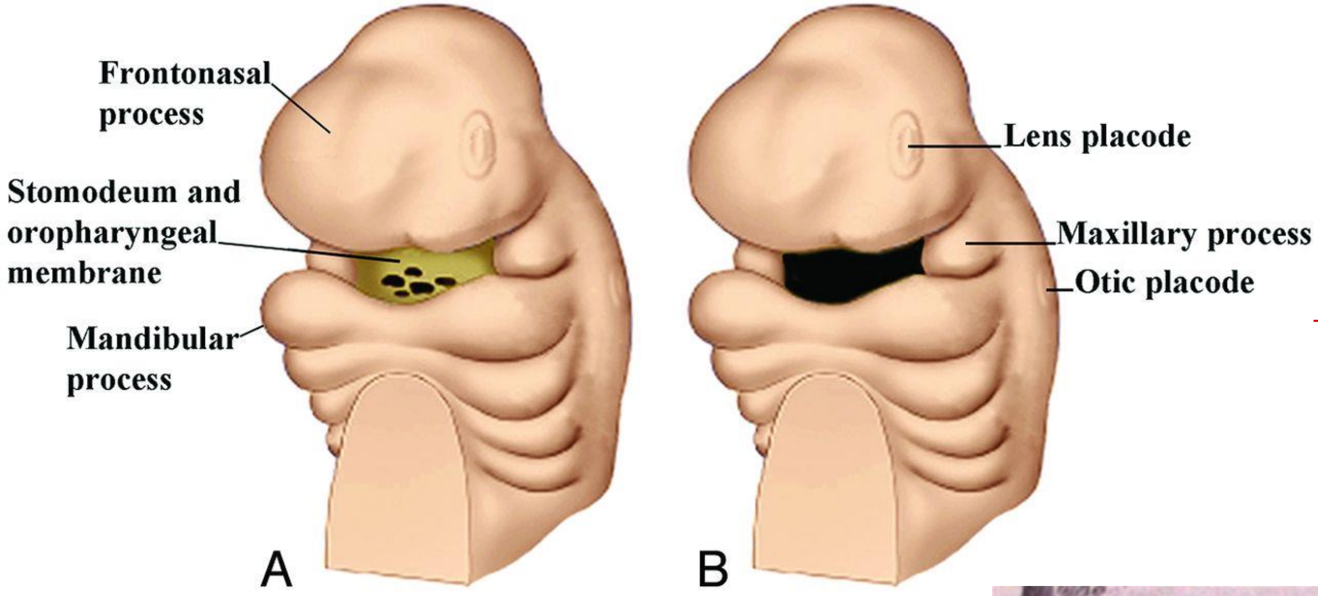
Eruption

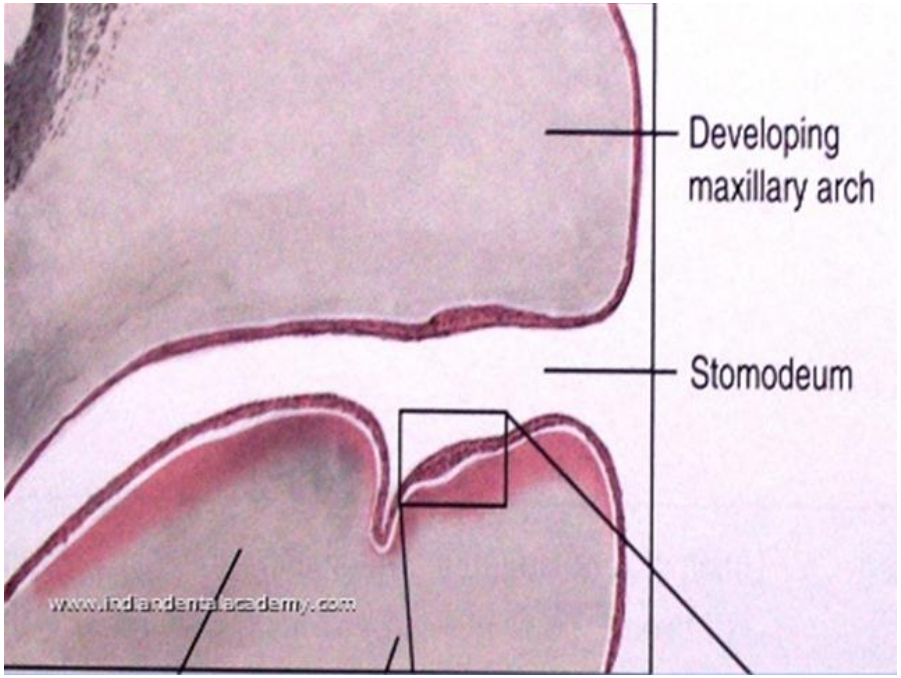
- Primitive oral cavity (stomodeum)lined by oral ectodermectoderm contacts mesenchyme
- Neural crest cells

“Neural crest cells migrate into the developing jaws and become ectomesenchyme.

Teeth need epithelium + ectomesenchyme working together.”


Cranial and sensory ganglia and nerves, adrenal medulla, ectomesenchyme bones and skull, dentin, periodontal ligament, alveolar bone
Neural crest
Posterior pituitary, pineal body, retina, central nervous system
Neuroectoderm
Urogenital system
Intermediate plate
Surface ectoderm
Epidermis, hair, nails, cutaneous glands, mammary glands, anterior pituitary gland, parenchyma of salivary gland, enamel of teeth, lens, inner ear
Paraxial
Muscles of trunk, skeleton (except skull), dermis of skin, connective tissue (mesenchyme)
Lateral plate
Connective tissue (mesenchyme), muscles of viscera, serous membranes of pleura, pericardium and peritoneum, blood and lymph cells, cardiovascular and lymphatic systems, spleen, adrenal cortex
Endoderm
Epithelial component of trachea, bronchi and lungs, epithelium of gastrointestinal tract, liver, pancreas, urinary bladder and urachus, epithelial component of pharynx, thyroid, tympanic cavity, pharyngotympanic tube, tonsils, and parathyroids

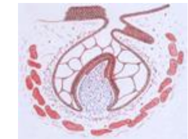

Frontonasal process
Stomodeum and oropharyngeal membrane
Mandibular process
A
Lens placode
Maxillary process
Otic placode
B


Developing maxillary arch
Stomodeum
www.indiandentalacademy.com

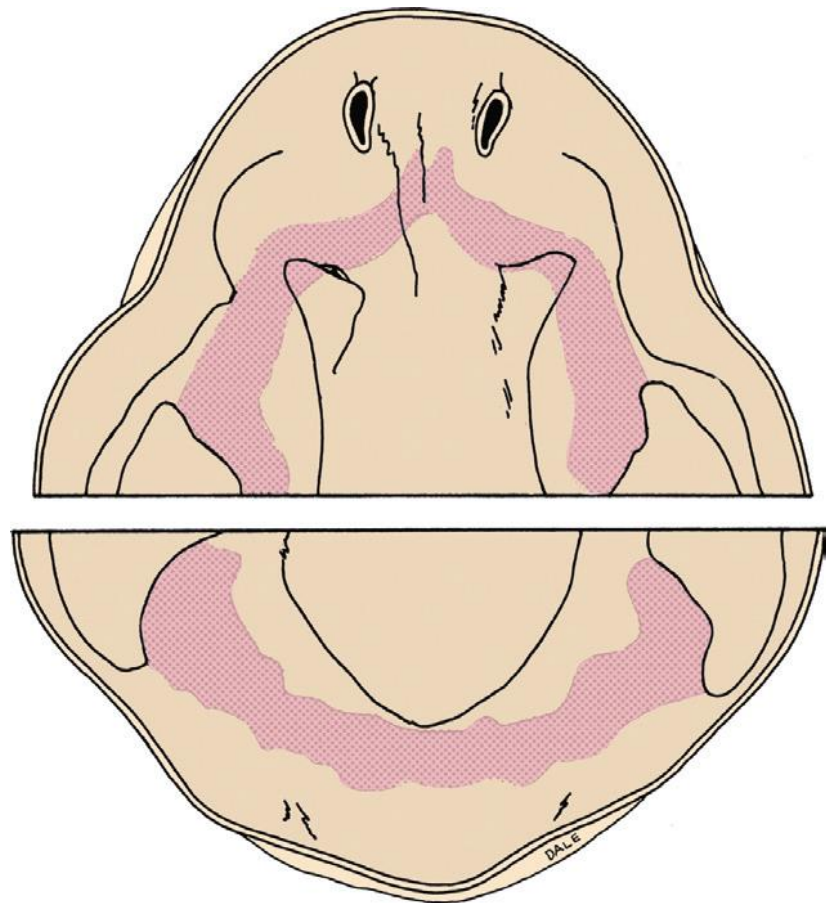
- Clinical correlation:

neural crest disturbance → craniofacial anomalies + dental anomalies

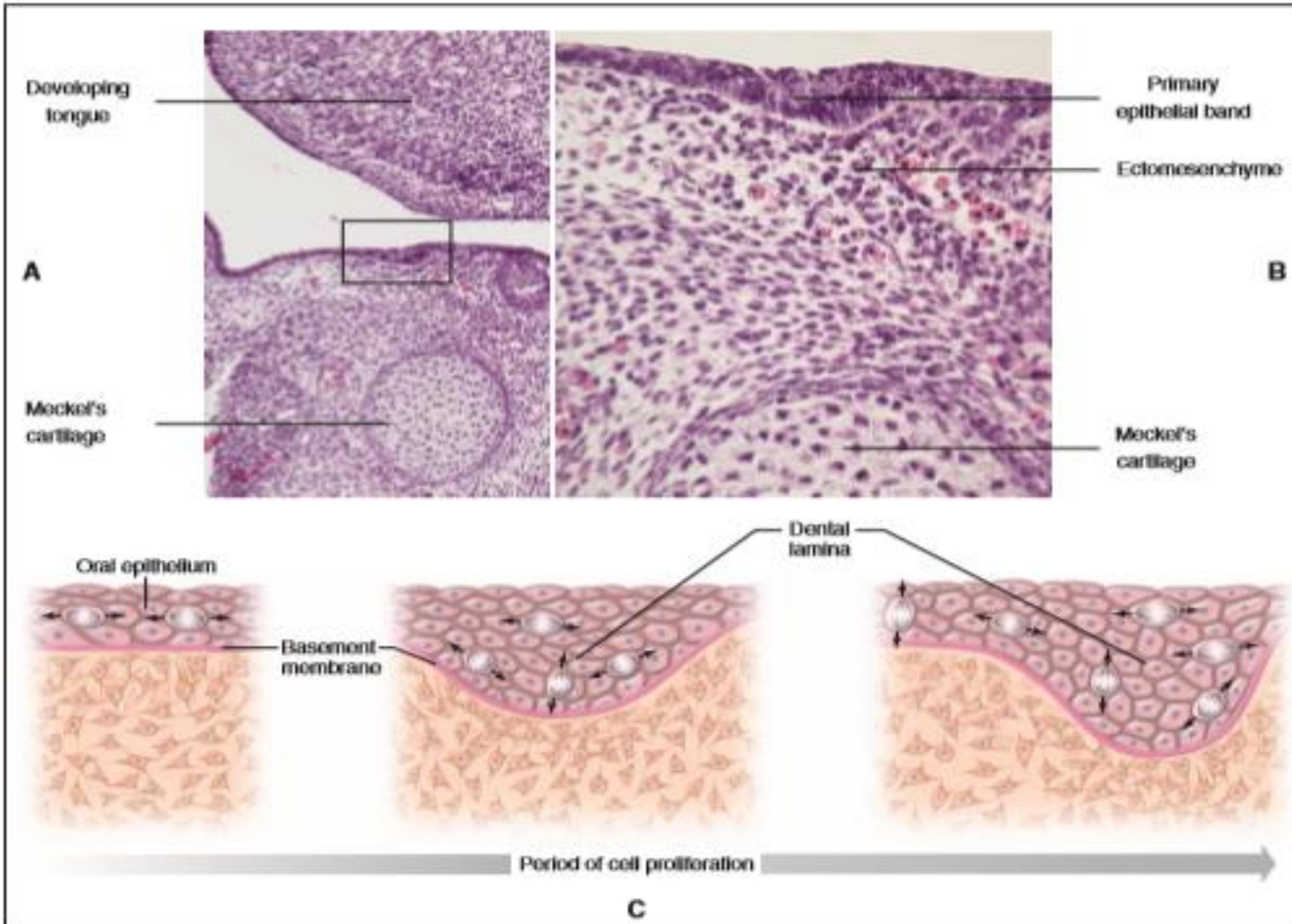
Primary epithelial band



- After 37 days of development
- Continuous band of thickened epithelium forms around the mouth in presumptive upper and lower jaw
- Horse-shoe shaped



- oral epithelium proliferates grows into ectomesenchyme forms dental lamina

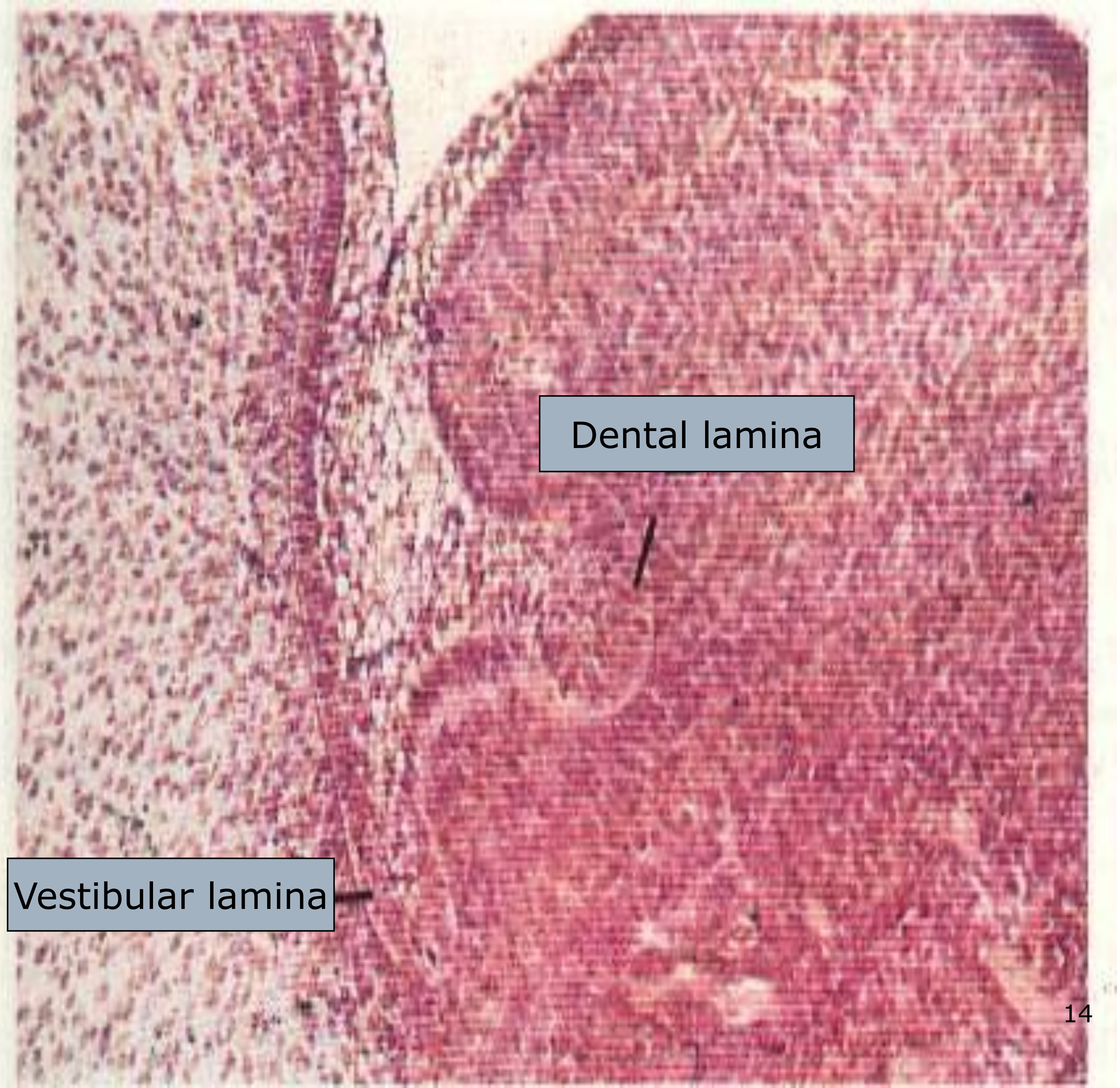

Developing tongue
Meckel's cartilage
A
Primary epithelial band
Ectomesenchyme
Meckel's cartilage
B
Dental lamina
Oral epithelium
Basement membrane
Period of cell proliferation
C

Primary epithelial band

- Correspond in position to the future dental arches
- Each band of epithelium, called the *primary epithelial band*, quickly gives rise to two subdivisions which ingrow into the underlying mesenchyme colonized by neural crest cells.
- Give rise to vestibular lamina & dental lamina

- A key feature of the initiation of tooth development is the formation of localized thickenings or placodes(dental placodes) within the primary epithelial bands.

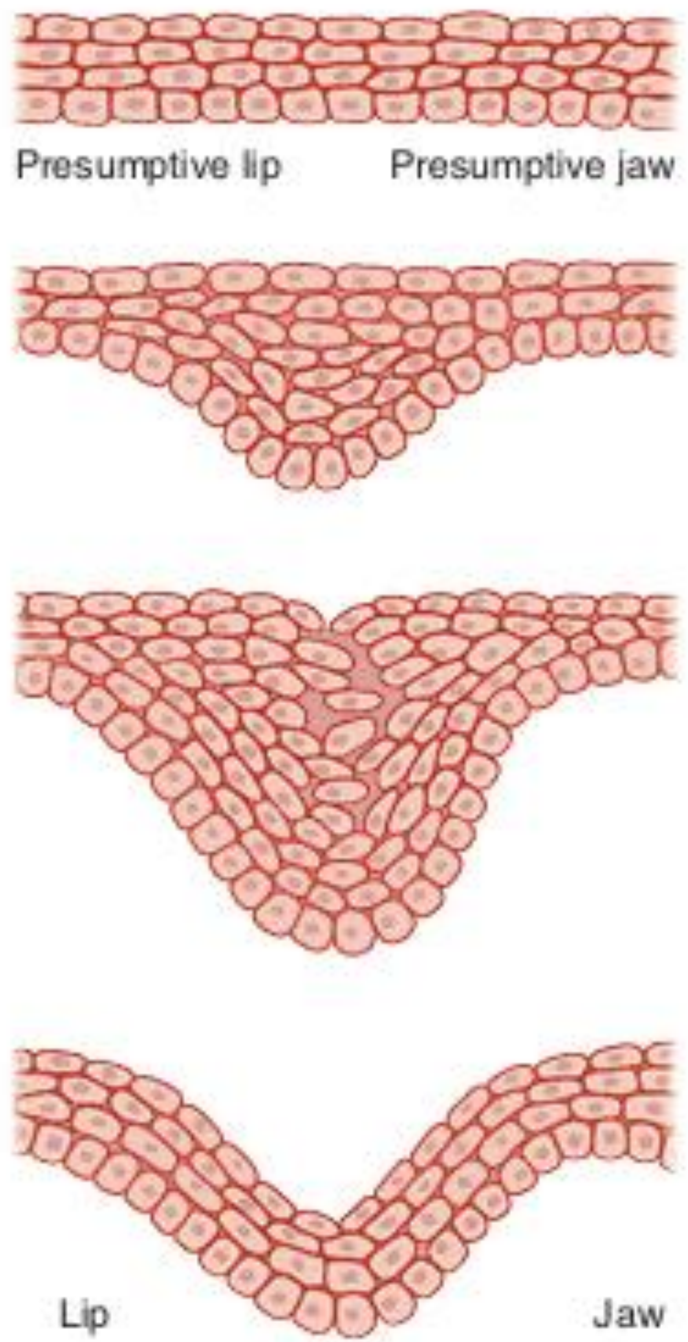

Primary epithelial band
ectomesenchyme


Dental lamina
Vestibular lamina

VESTIBUAR LAMINA

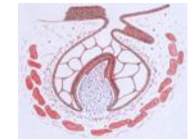
- Vestibule is formed as a result of the proliferation of the vestibular lamina into the ectomesenchyme

- Cells rapidly enlarge and then degenerate



Presumptive lip
Presumptive jaw
Lip
Jaw

DENTAL LAMINA



- continued & localized proliferative activity

- Formation of a series of epithelial out growth into the ectomesenchyme at sites, corresponding to the position of future deciduous teeth

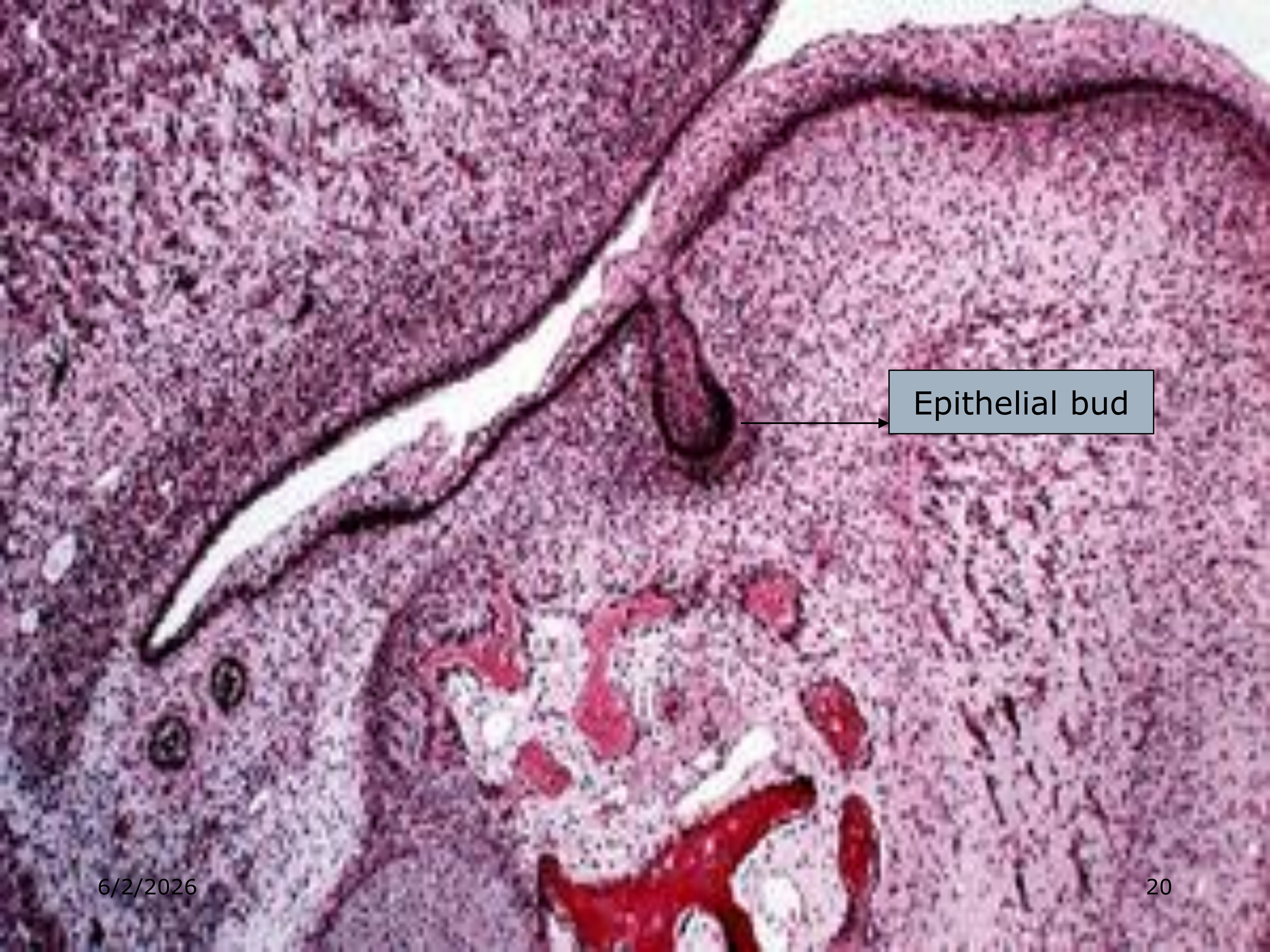
- Ectomesenchymal cells accumulate around these outgrowths

- Bud stage → cap stage → bell stage

BUD STAGE

- Represented by first epithelial incursion into the ectomesenchyme of the jaw
- Epithelial cells show little change in shape & function
- Supporting ectomesenchymal cells are packed closely beneath & around the epithelial bud

- As the epithelial bud continues to proliferate into the ectomesenchyme, cellular density increases immediately adjacent to the epithelial outgrowth. This process is classically referred to as a *condensation of the ectomesenchyme.*


Epithelial bud

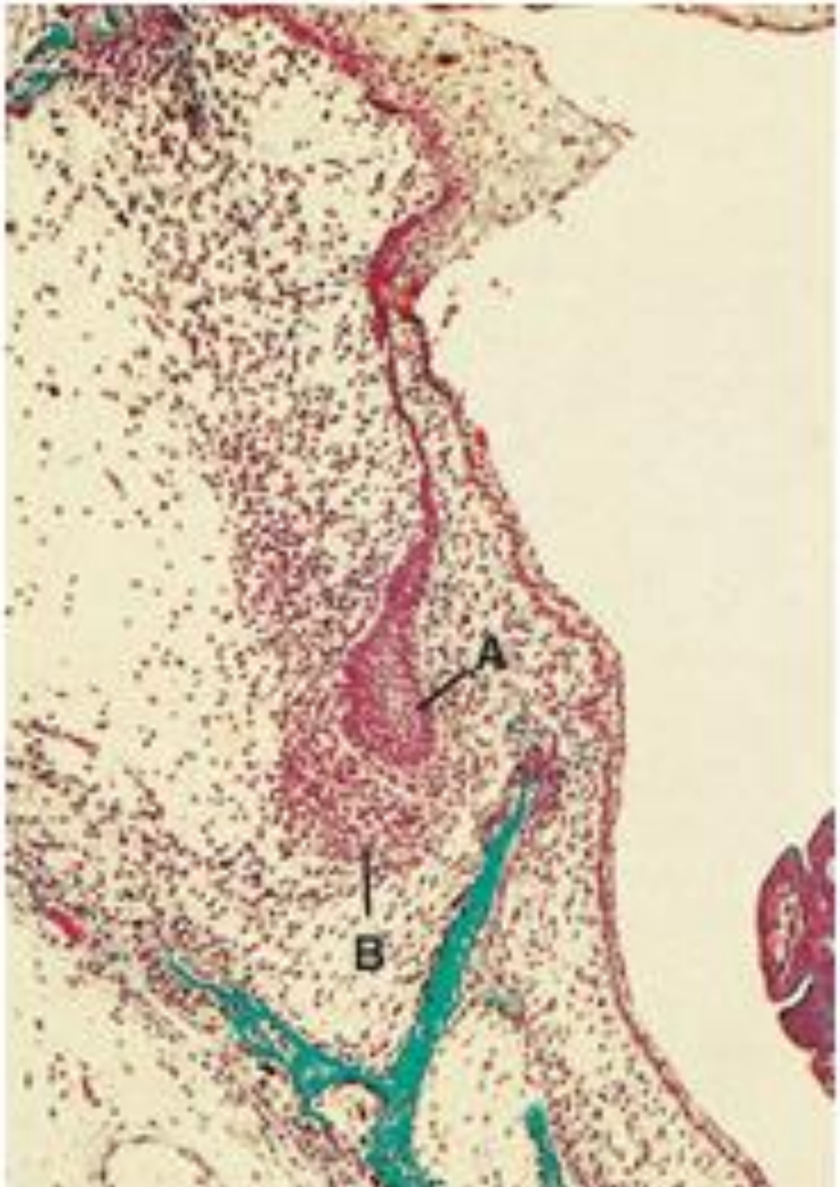


Fig. 21.6 Bud stage of tooth development. A = enamel organ; B = mesenchymal condensation (Masson's trichrome; ×60).

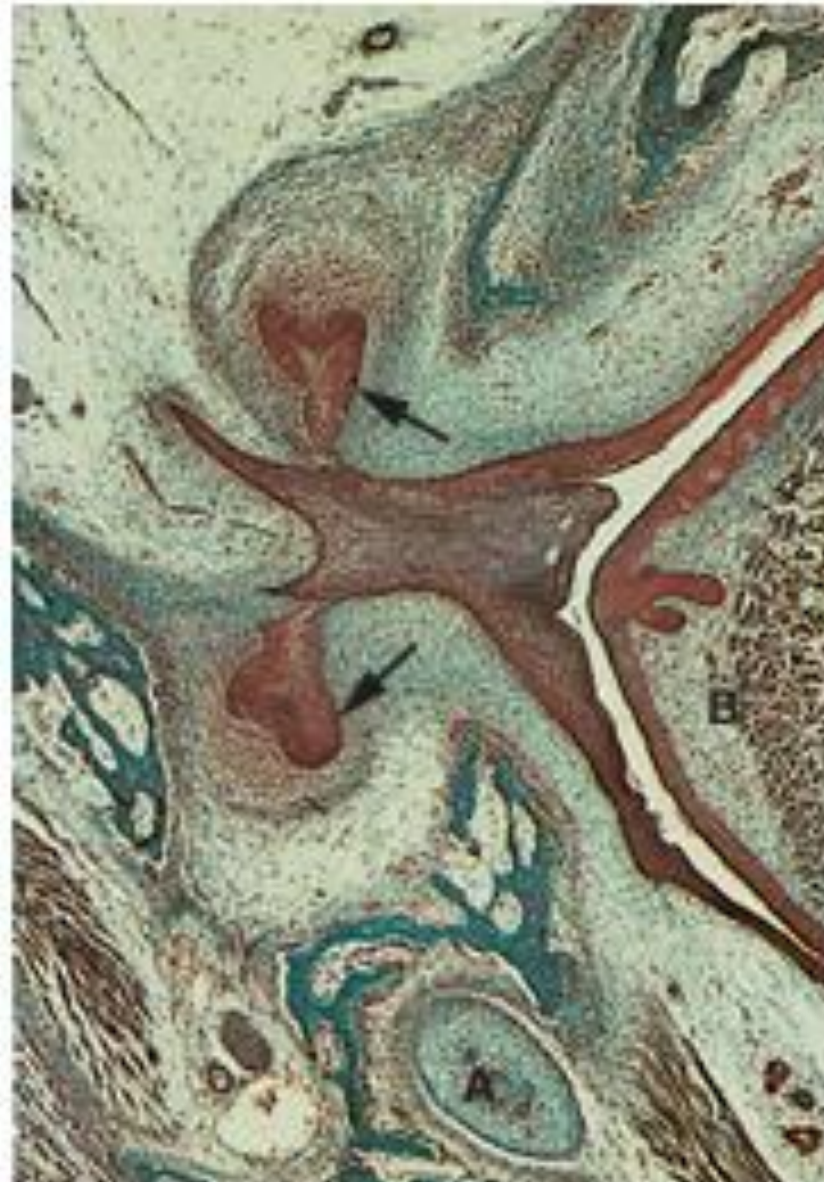
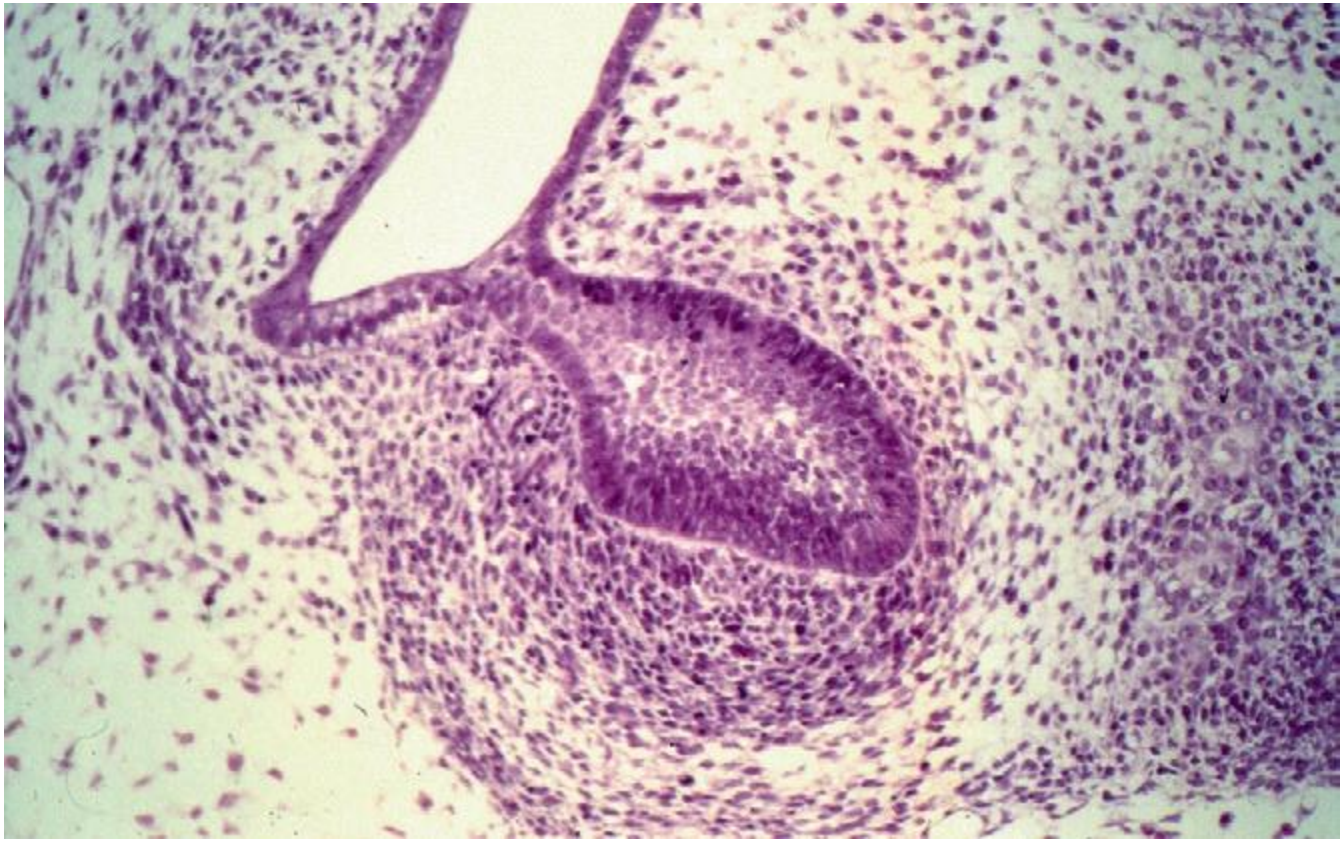


Fig. 21.7 Early cap stage of tooth development (arrows). A = Meckel's cartilage; B = developing tongue (Masson's trichrome; ×32).

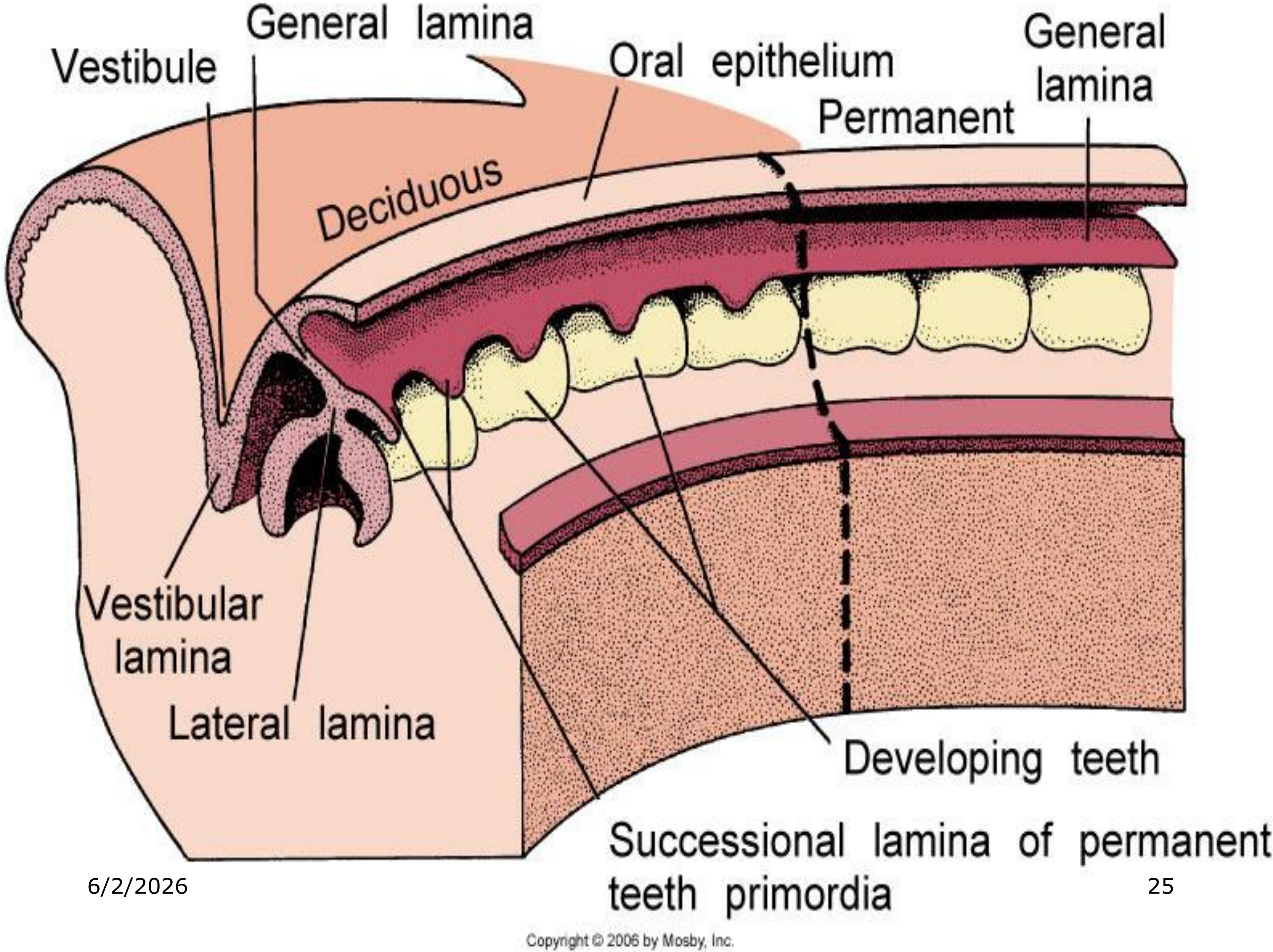
Cap Stage

- Epithelial bud continues to proliferate into the ectomesenchyme.
- Cellular density increases immediately adjacent to epithelial outgrowth.



Cap Stage

- CONDENSATION of ectomesenchyme results from local grouping of cells that failed to produce extracellular matrix and are not separated from each other.
- Tooth bud drags along part of dental lamina, developing tooth tethered to the dental lamina by an extension known as Lateral lamina.


General lamina
Vestibule
Oral epithelium
General
lamina
Permanent
Deciduous
Vestibular
lamina
Lateral lamina
Developing teeth
Successional lamina of permanent
teeth primordia
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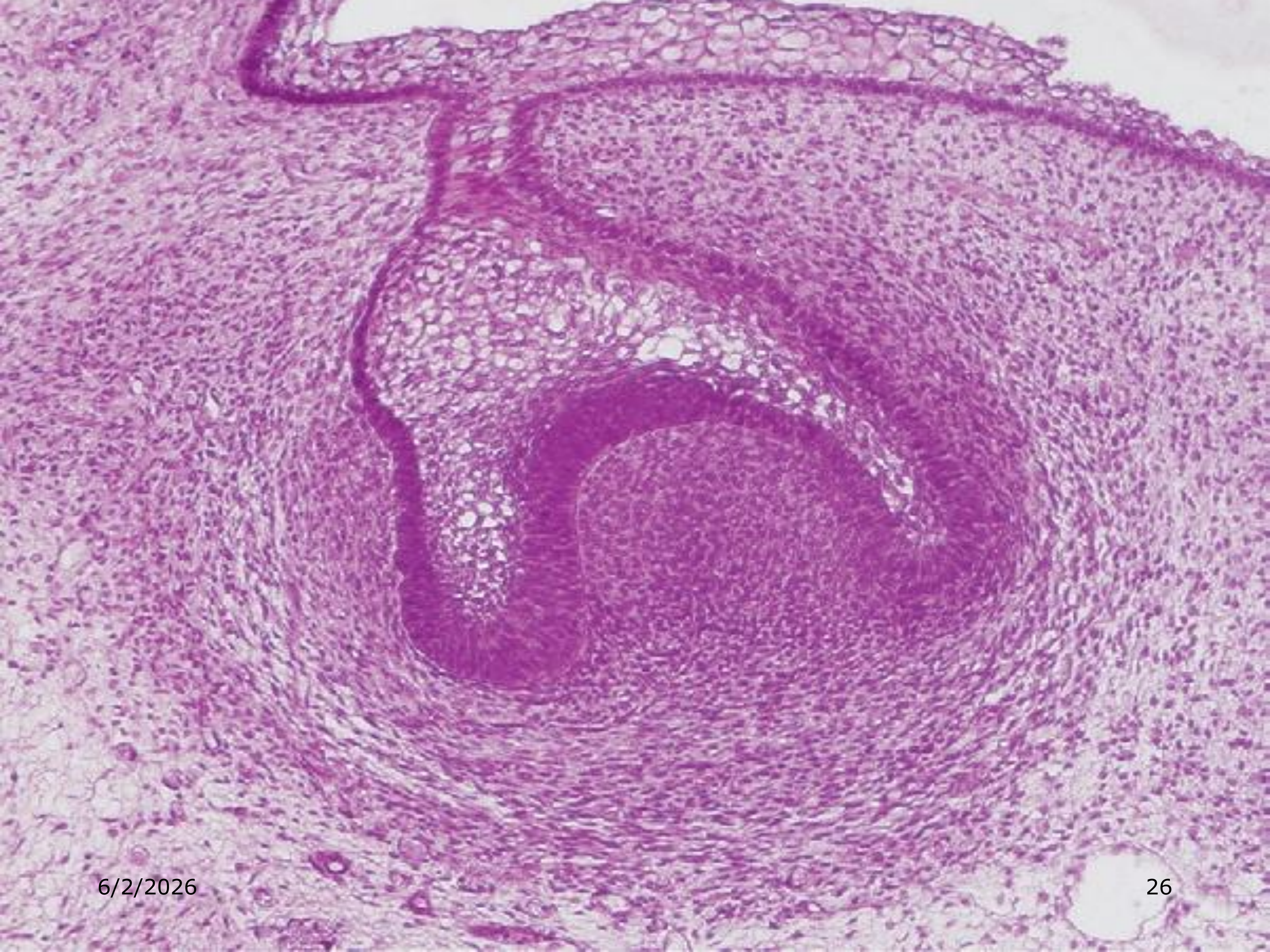


Cap Stage

- Formative elements of tooth and its supporting tissues can be identified.
- DENTAL ORGAN or ENAMEL ORGAN: Epithelial outgrowth, which superficially resembles a cap sitting on a ball of condensed ectomesenchyme

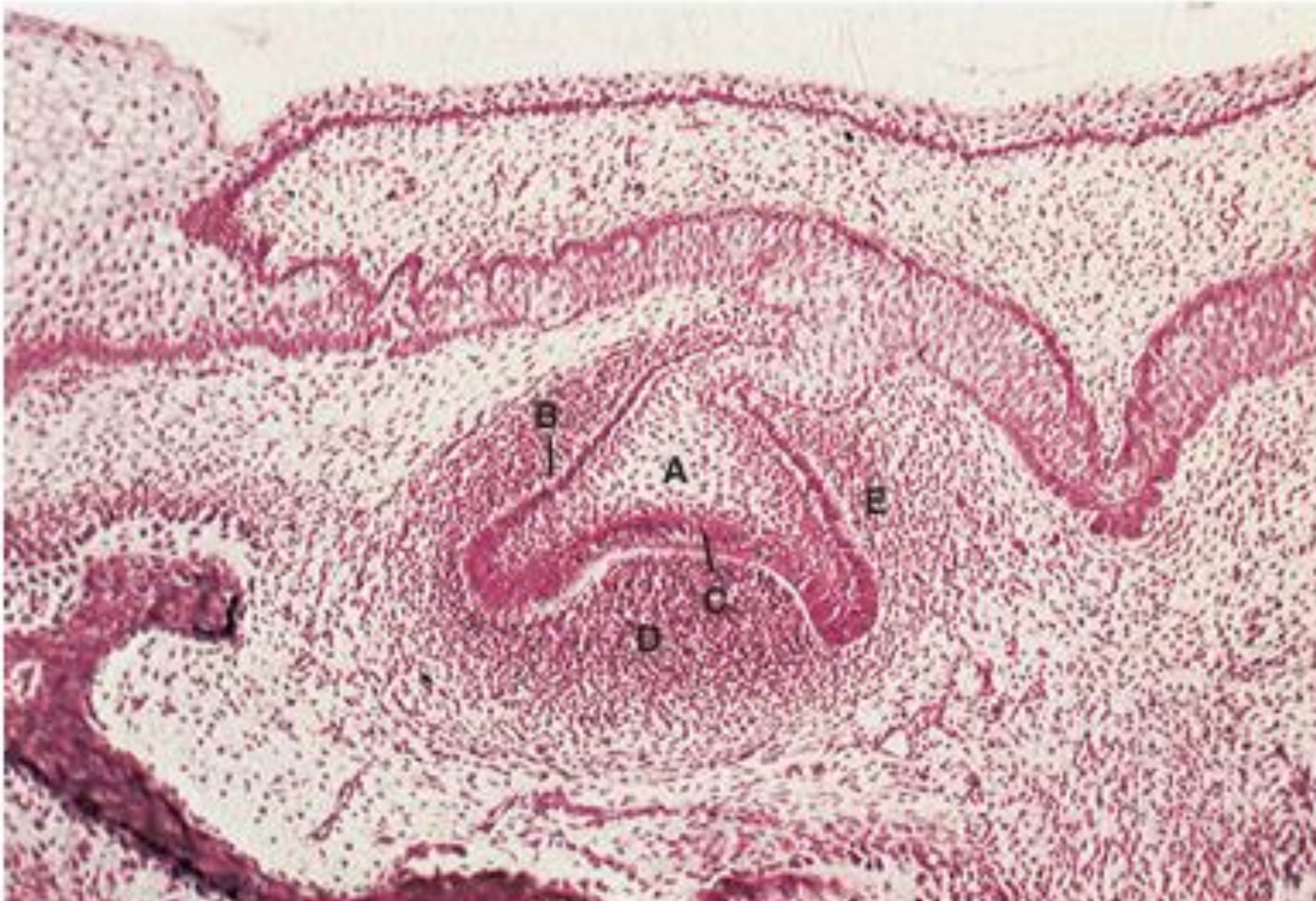


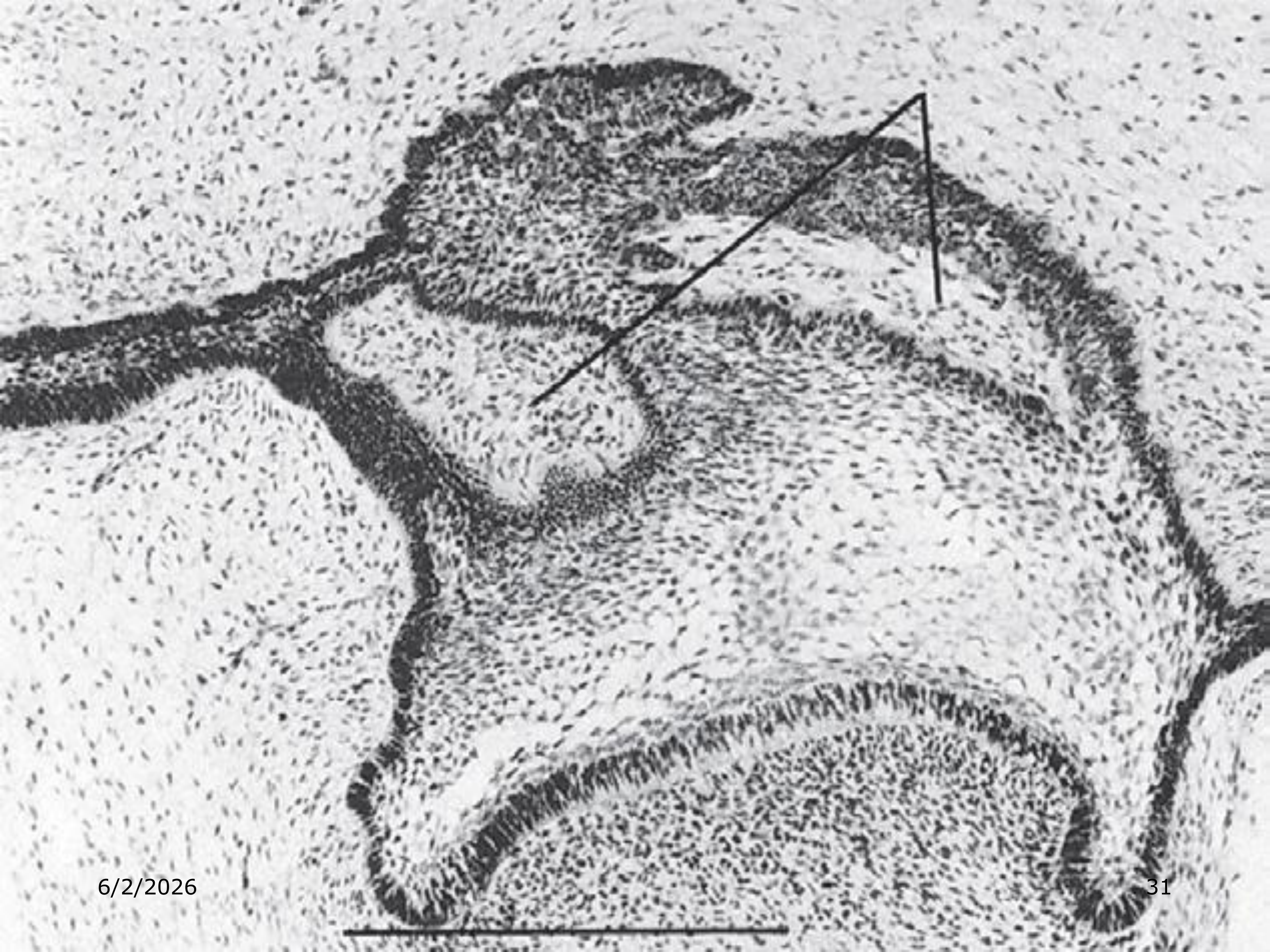
Fig. 21.8 Late cap stage of tooth development. A = stellate reticulum; B = external enamel epithelium; C = internal enamel epithelium; D = dental papilla; E = dental follicle (H & E; ×75).

Cap Stage

- DENTAL PAPILLA: Condensed ectomesenchymal cells forms the dentine and pulp.
- DENTAL FOLLICLE: Condensed ectomesenchyme limiting the dental papilla and encapsulating the enamel organ forms supporting tissues of the tooth.
- The enamel organ, dental papilla, and dental follicle together constitute the dental organ or tooth germ.

Enamel Niche

- The enamel niche is seen where the tooth germ appears to have a double attachment to the dental lamina (the lateral and medial enamel strands).
- These strands enclose the enamel niche, which appears as a funnel-shaped depression containing connective tissue.
- The functional significance of the enamel niche is unknown.



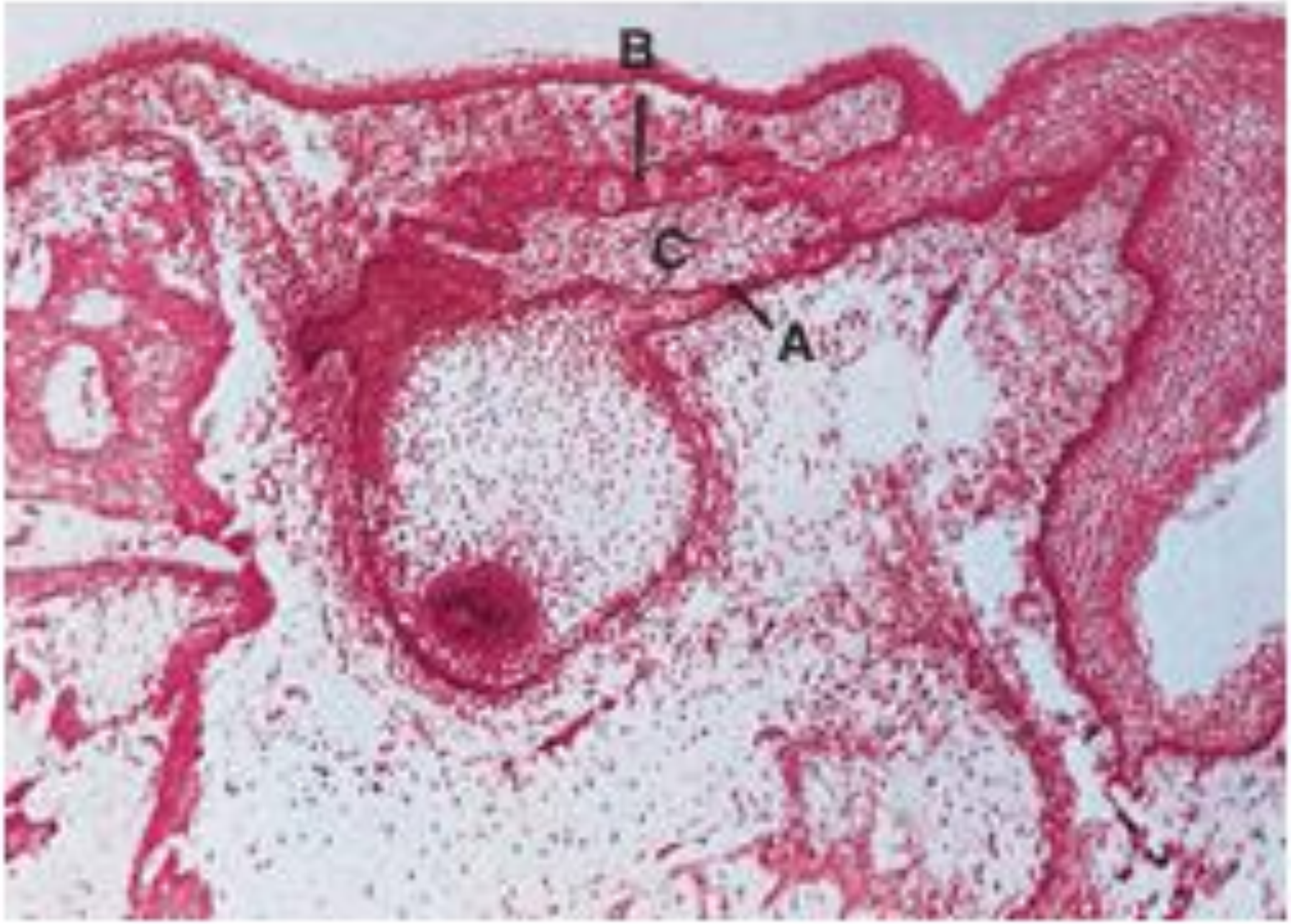
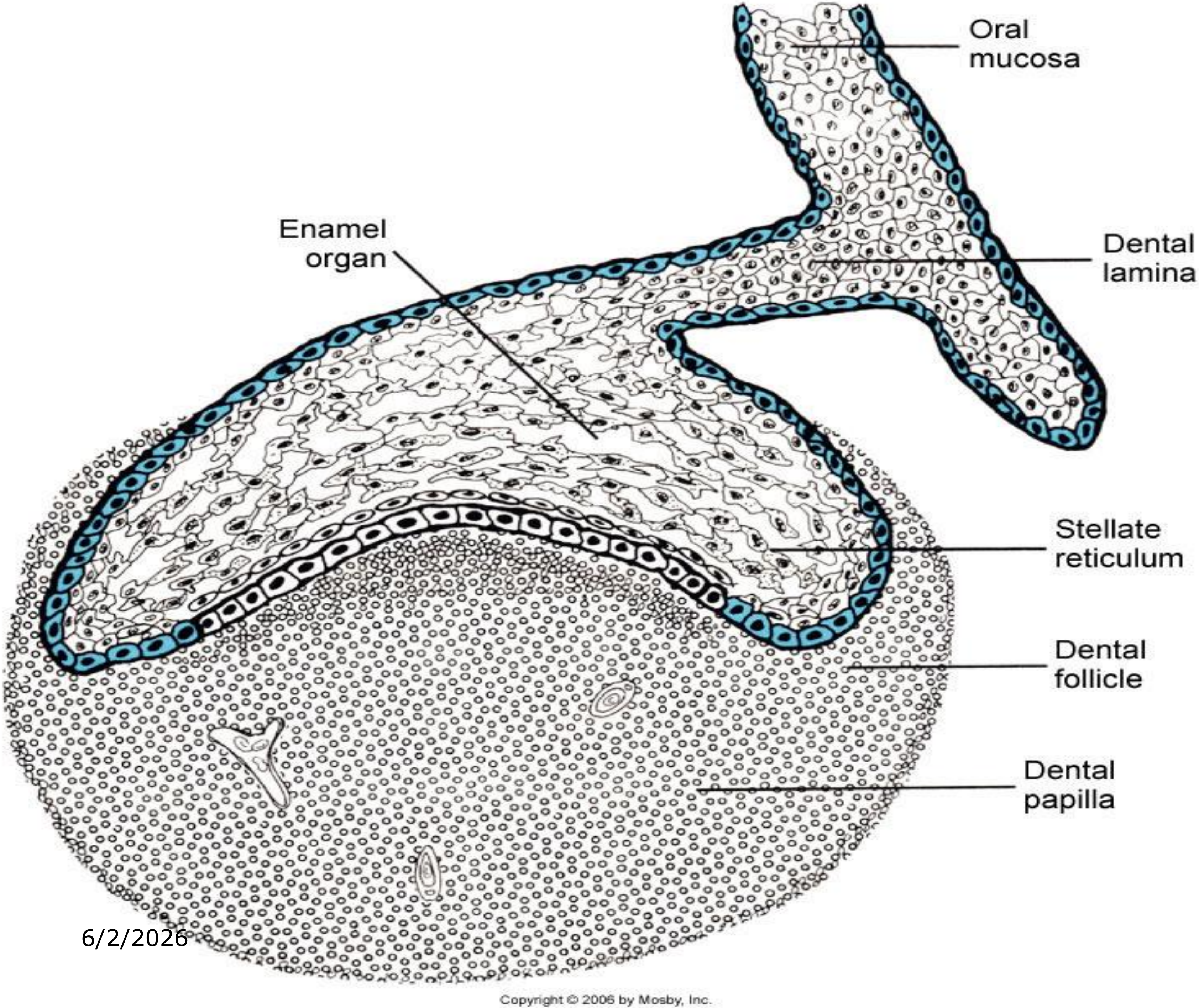


Fig. 21.22 The enamel niche (C). A = lateral enamel strand; B = medial enamel strand (H & E; ×40).


Oral
mucosa
Enamel
organ
Dental
lamina
Stellate
reticulum
Dental
follicle
Dental
papilla
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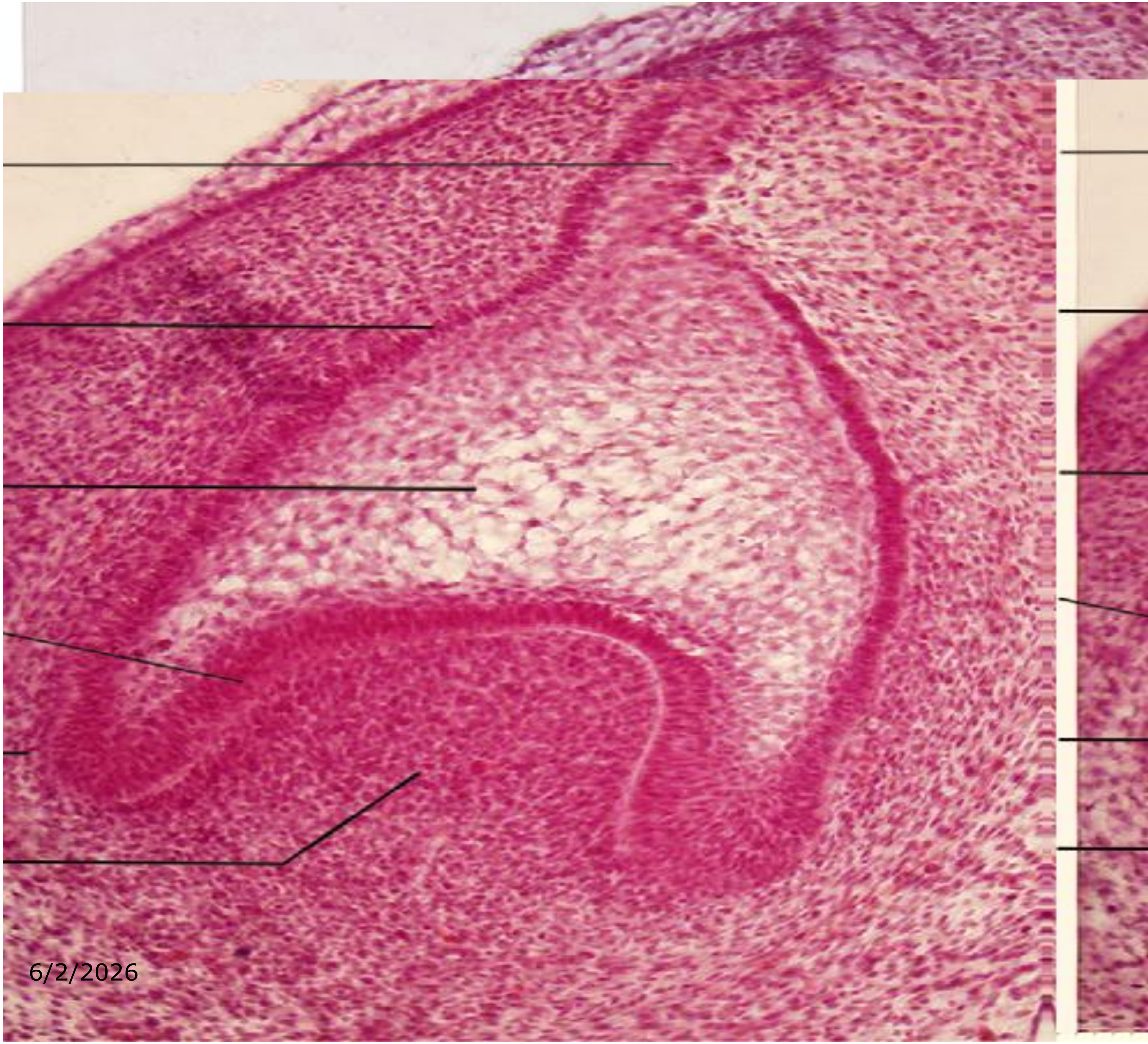
Cap Stage

- Cells in centre of enamel organ synthesize and secrete glycosaminoglycans into the extracellular compartment between the epithelial cells.
- Glycosaminoglycans are hydrophilic and so pull water into the enamel organ.
- Central cells are forced apart due to increase in fluid but retain their desmosomal contacts and become star shaped.
- Centre of enamel organ is termed as STELLATE RETICULUM



Enamel knots

- This is a localised, nonproliferating cluster of cells at the centre of the inner dental epithelium, which is central to establishing the form of the tooth crown.

- The enamel knot precursor cells can be detected first at the tip of the tooth buds

- **Enamel knot represents an organizational center that orchestrates cuspal morphogenesis.**

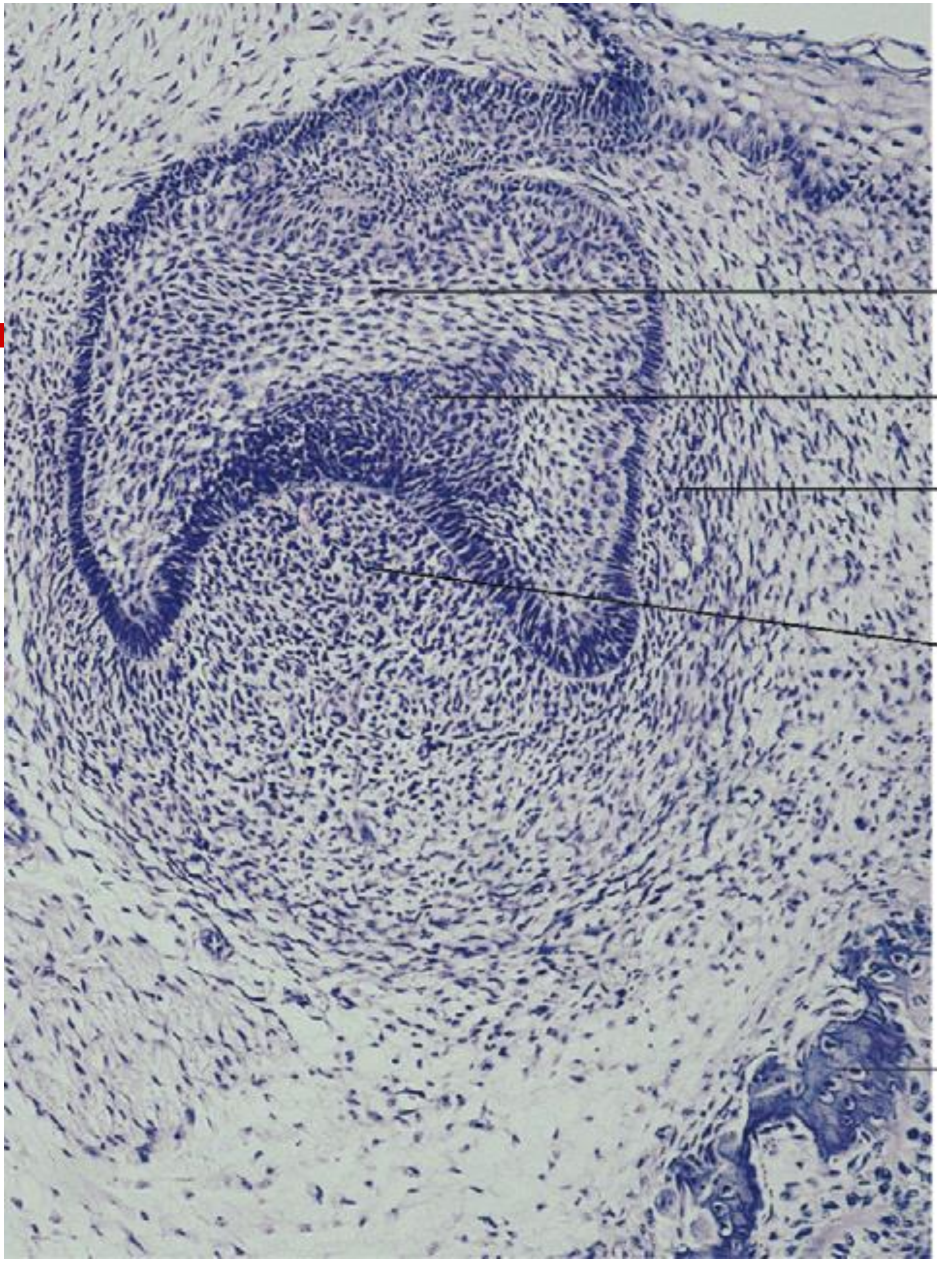


Fig. 21.18 The enamel knot (A) (H & E; ×120).

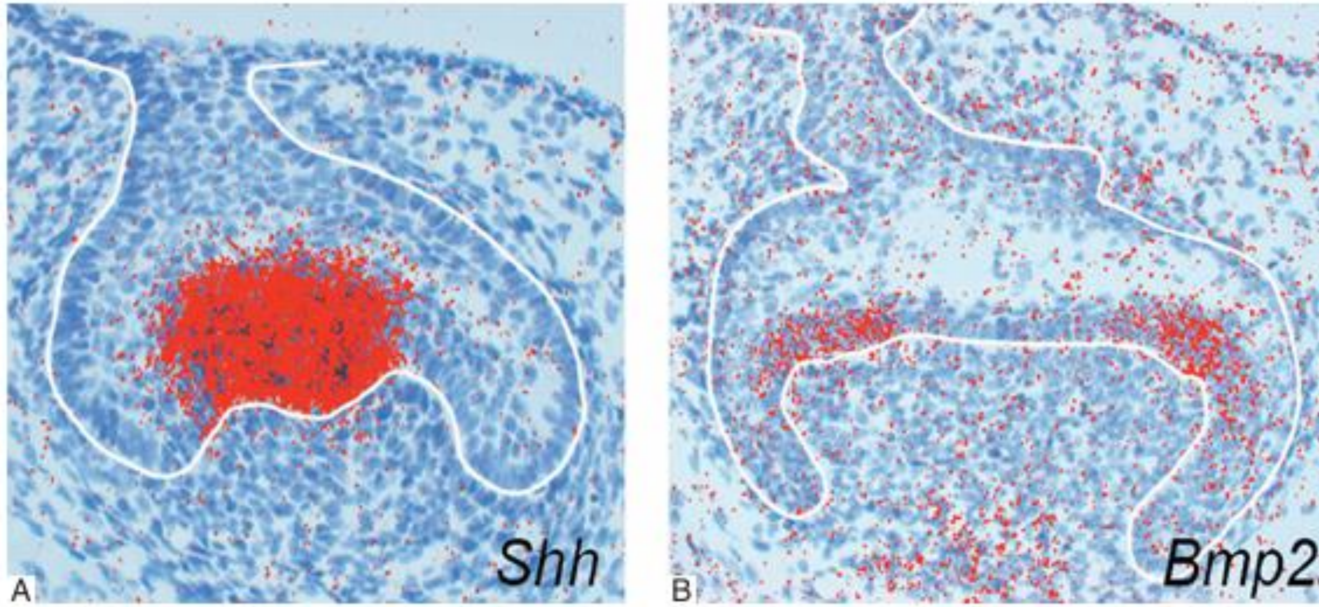


Fig. 21.19 (A) Staining for Shh (sonic hedgehog – red stain) in the primary enamel knot of a molar tooth by *in situ* hybridization (×140). (B) Staining for BMP-2 (red stain) in the secondary enamel knots of a molar tooth by *in situ* hybridization (×140). Courtesy Dr M Jussila.

Enamel cord

- The enamel cord is a strand of cells seen at the early bell stage of development that extends from the stratum intermedium into the stellate reticulum.
- When present, the enamel cord overlies the incisal margin of a tooth or the apex of the first cusp to develop (primary cusp).
- It has been suggested that the enamel cord may be involved in the process by which the cap stage is transformed into the bell stage (acting as a mechanical tie) or that it is a focus for the origin of stellate reticulum cells

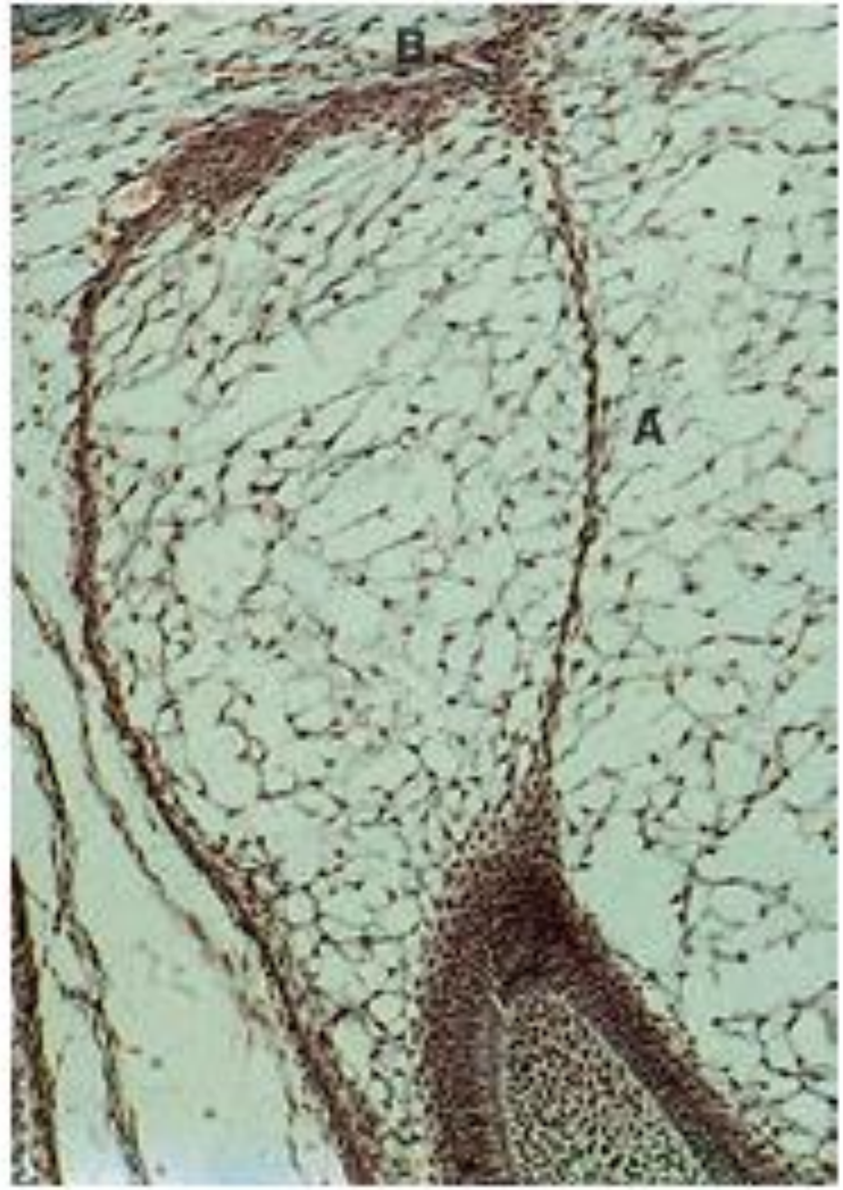


Fig. 21.21 The enamel cord (A). B = enamel navel (Masson's trichrome; ×120).

- When it completely divides the stellate reticulum into two parts, reaching the external enamel epithelium, it is termed the enamel septum.
- Where the enamel cord meets the external enamel epithelium, a small invagination termed the enamel navel may be seen.

BELL STAGE

- so called because the enamel organ comes to resemble a bell as the undersurface of the epithelial cap deepens.

- During this stage, the tooth crown assumes its final shape (morphodifferentiation), and the cells that will be making the hard tissues of the crown (ameloblasts and odontoblasts) acquire their distinctive phenotype (histodifferentiation)

BELL STAGE

- At the periphery of enamel organ the cells assume a low cuboidal shape and forms the outer or external dental epithelium.

- Cells bordering the dental papilla assume a short columnar shape and forms the inner or internal dental epithelium. they have high glycogen content.

BELL STAGE

- Inner epithelium begins at the point where the outer epithelium bends to form the concavity into which the cells of dental papilla accumulate.
- Cervical loop or the zone of reflexion is the area where internal and external enamel epithelium meets.

Bell stage

A-external enamel epithelium

B-internal enamel epithelium lies the cervical loop

C- Stellate reticulum

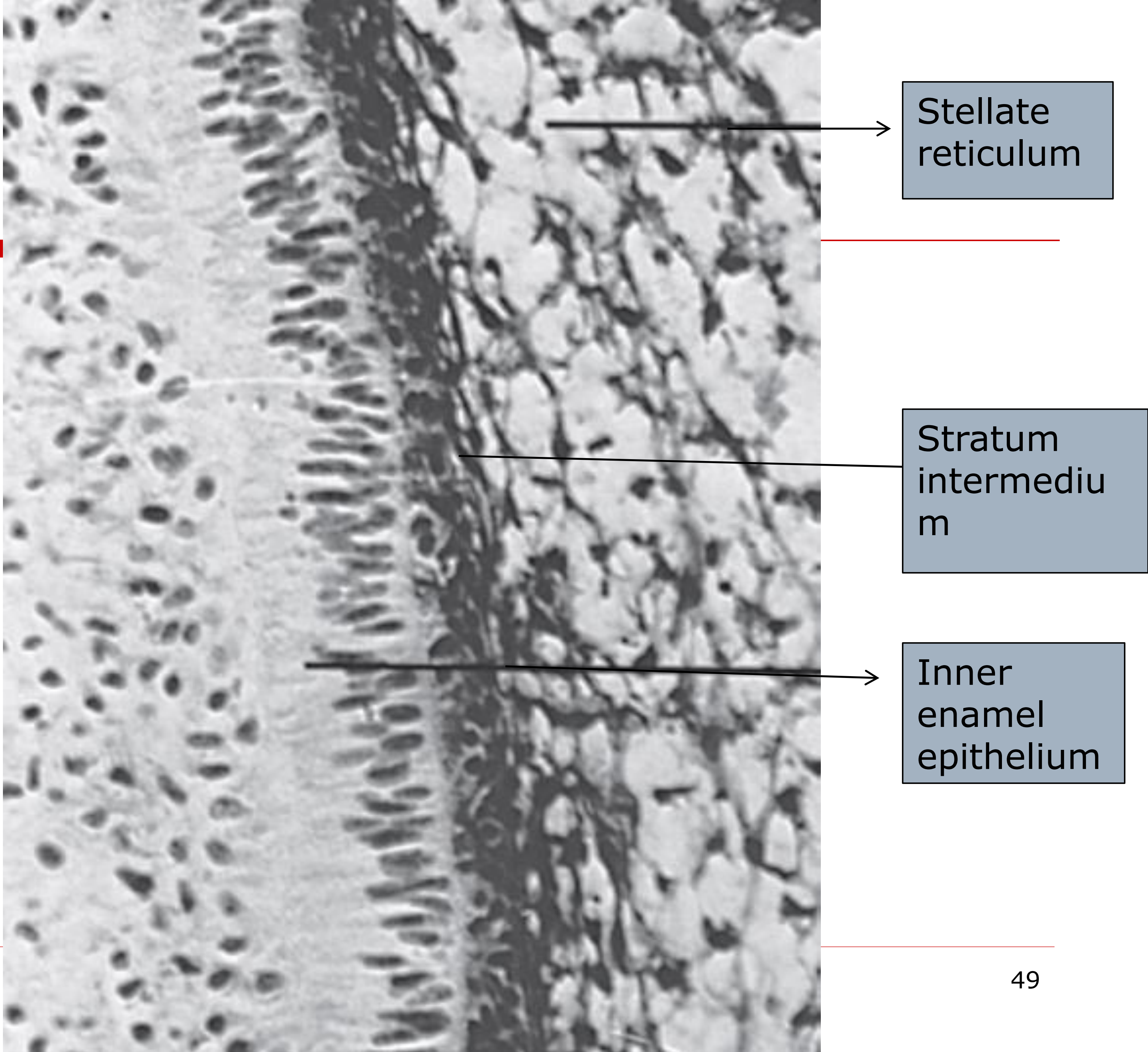
D-enamel cord

BELL STAGE

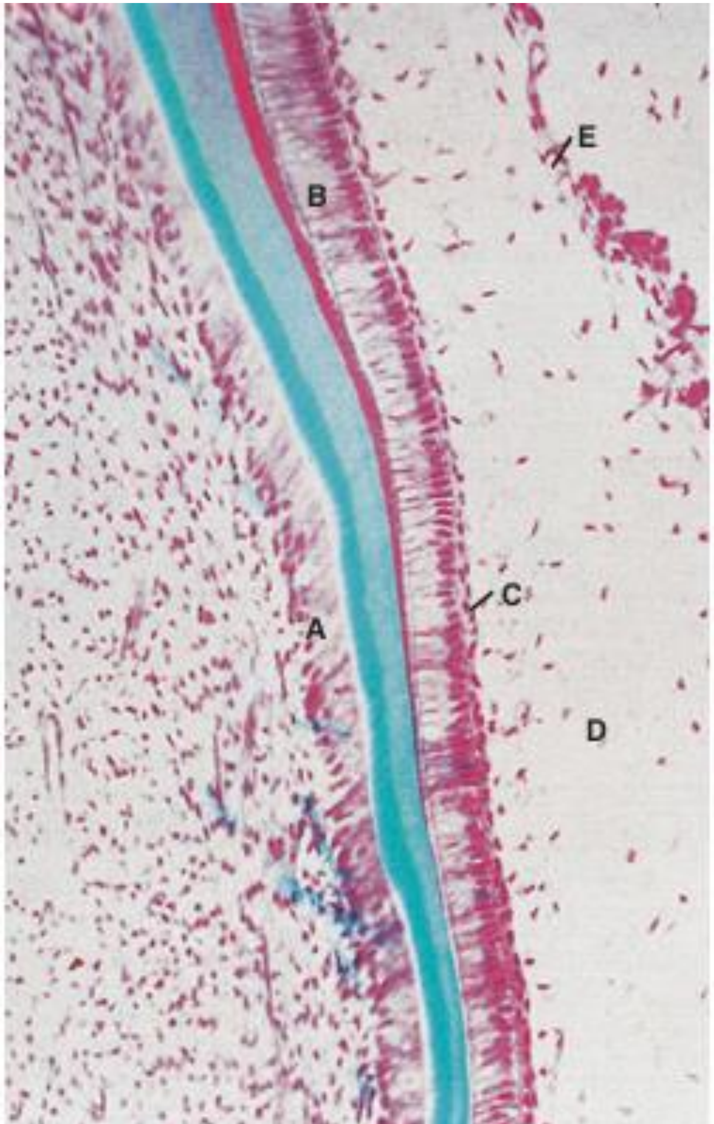
- At the cervical loop area the cells continue to divide until the tooth crown attains its full size.
- After crown formation gives rise to epithelial component of root formation.

BELL STAGE

- Some epithelial cells between the inner dental epithelium and Stellate reticulum differentiate into a layer called the stratum intermedium.
- Characterized by High activity of enzyme alkaline phosphatase.
- Along with cells of inner dental epithelium forms enamel.


Stellate reticulum
Stratum intermedium
Inner enamel epithelium

Late bell stage

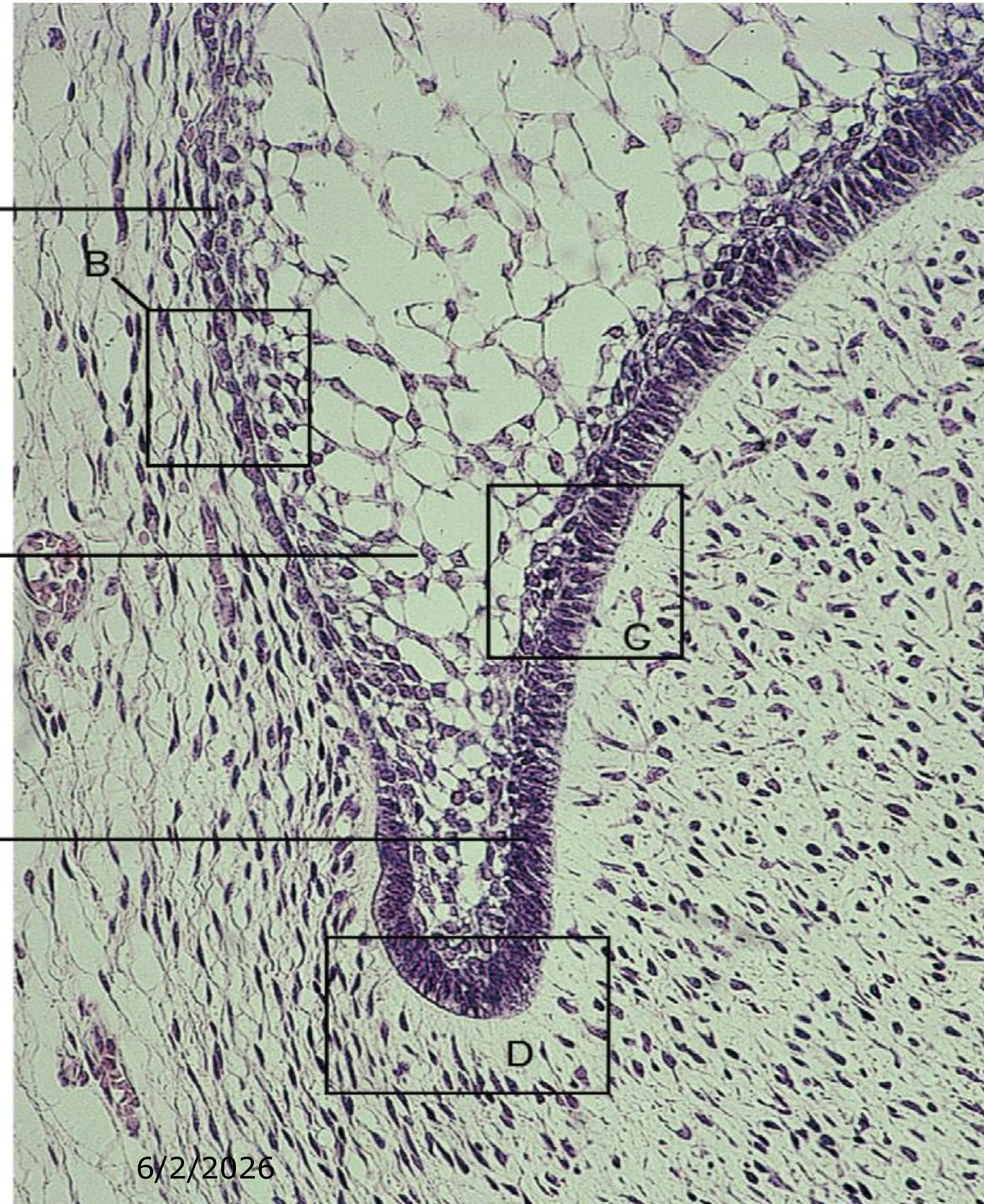


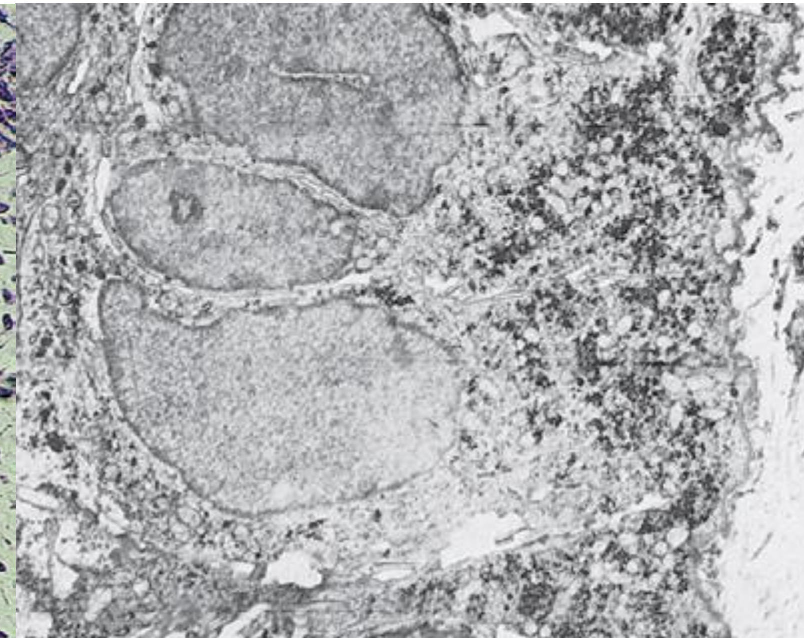
- A-odontoblasts
- B-ameloblasts
- C-stratum intermedium
- D-Stellate reticulum
- E- External enamel epith

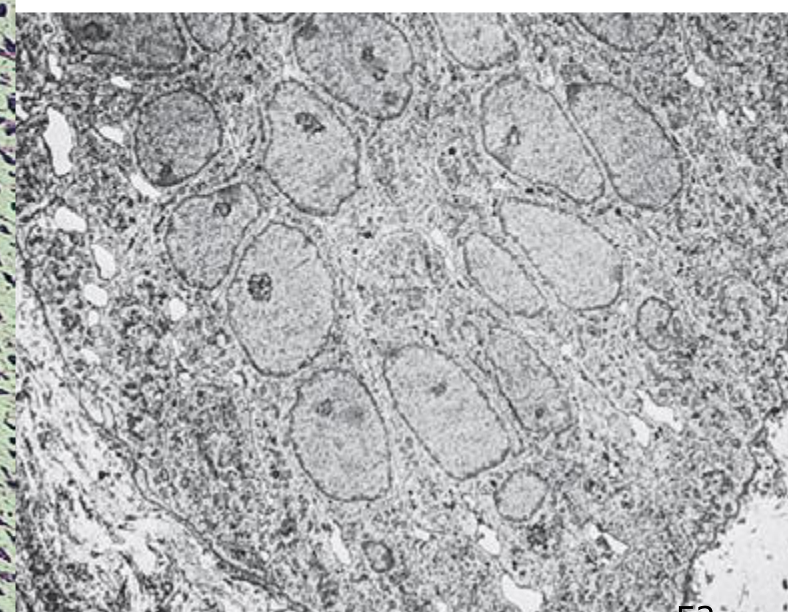
BELL STAGE

DENTAL PAPILLA:

- Separated from dental organ by a basement membrane.
- From basement membrane a mass of fine aperiodic fibrils extend into an acellular zone.
- Fibrils correspond to lamina reticularis, site where first secreted enamel matrix proteins accumulate.


B
C
D





BELL STAGE

- Dental papilla contains undifferentiated mesenchymal cells and few fine scattered collagen fibers.

- At the bell stage the dental papilla is referred as the tooth pulp when the first calcified matrix appears at the cuspal tip.

BELL STAGE

DENTAL FOLLICLE:

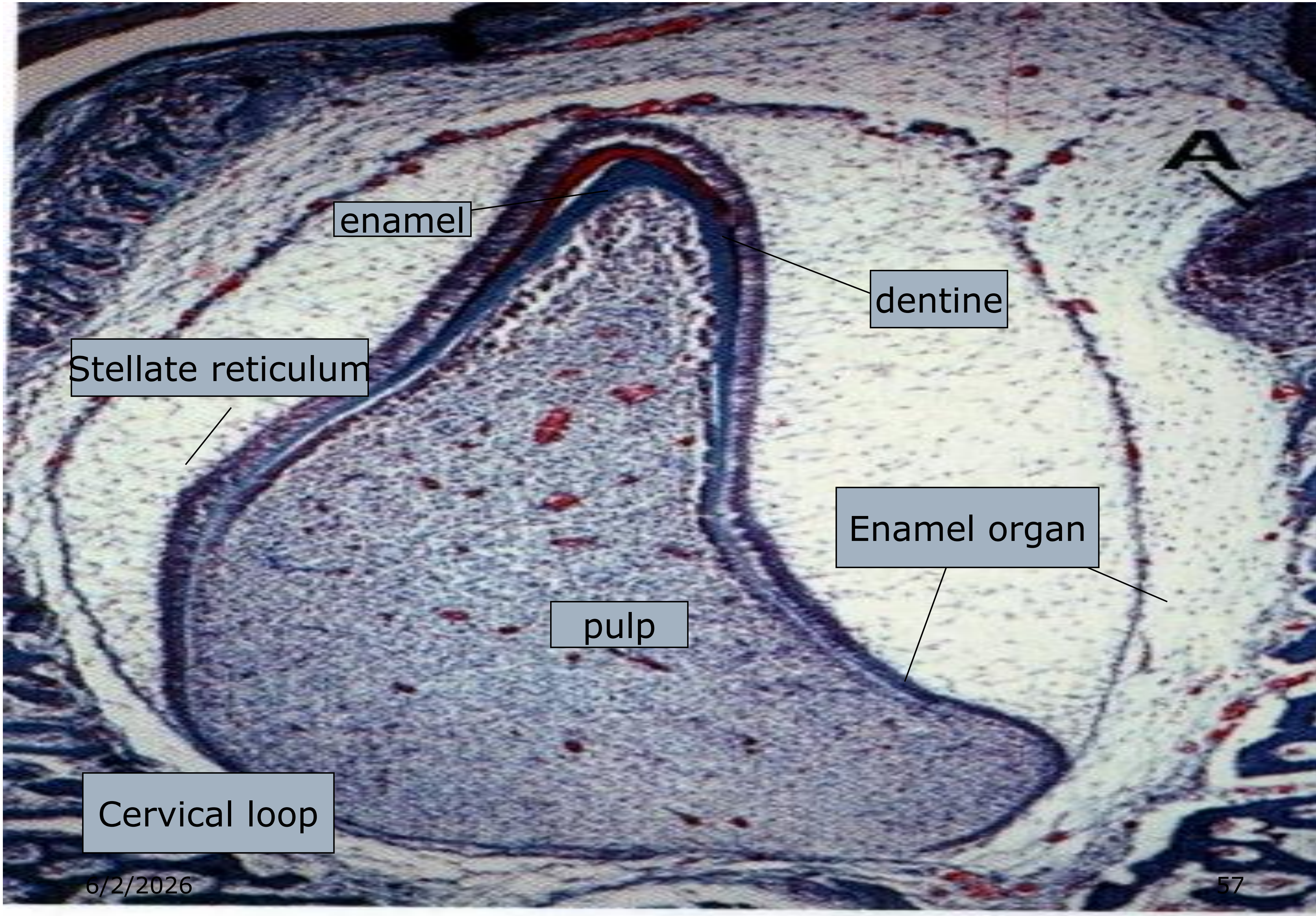
- More collagen fibrils occupying the extracellular spaces between the follicular fibroblast.

- Oriented circularly around the dental organ and dental papilla.

- Two other important events occur during the bell stage.

- First, the dental lamina (and the lateral lamina) joining the tooth germ to the oral epithelium fragments, eventually separating the developing tooth from the oral epithelium.

- Second, the inner enamel epithelium completes its folding, making it possible to recognize the shape of the future crown pattern of the tooth


enamel
dentine
Stellate reticulum
Enamel organ
pulp
Cervical loop
A

BELL STAGE

DENTAL LAMINA:

- Breaks and separate the developing tooth from oral epithelium.
- Most cells degenerates, but some may persist known as **Epithelial pearls**.
- May form eruption cyst and delays eruption, or odontomata or supernumery teeth.
- For eruption, tooth penetrate the lining epithelium & cause break in epithelium
- Integrity is reestablished by formation of a special seal around tooth, the **junctional epithelium.**

NERVE SUPPLY DURING EARLY DEVELOPMENT

- During the bud to cap stage nerve fiber approach and form a rich plexus in the dental follicle.
- Do not enter dental papilla until dentinogenesis begins.
- Sensory innervation to the future PDL and pulp.
- No ANS supply.

THANK YOU